

DCE: UCI

Date Given: April 10, 2023

Due Date:

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There are 5 problems in this homework assignment. Write a program in Python to solve these problems.

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1. Write a program which will find all such numbers which are divisible by 7 but are not a multiple of 5, between 2000 and 3200 (both included). Add these numbers in a list.

Print the count of such numbers. Print your answer with 10 numbers in each line. Your output should look as follows.

```
Total Number of elements = 138
```

```
0 . 2002, 1 . 2009, 2 . 2016, 3 . 2023, 4 . 2037, 5 . 2044, 6 . 2051, 7 . 2058, 8 . 2072, 9 . 2079,
10 . 2086, 11 . 2093, 12 . 2107, 13 . 2114, 14 . 2121, 15 . 2128, 16 . 2142, 17 . 2149, 18 . 2156, 19 . 2163,
20 . 2177, 21 . 2184, 22 . 2191, 23 . 2198, 24 . 2212, 25 . 2219, 26 . 2226, 27 . 2233, 28 . 2247, 29 . 2254,
30 . 2261, 31 . 2268, 32 . 2282, 33 . 2289, 34 . 2296, 35 . 2303, 36 . 2317, 37 . 2324, 38 . 2331, 39 . 2338,
40 . 2352, 41 . 2359, 42 . 2366, 43 . 2373, 44 . 2387, 45 . 2394, 46 . 2401, 47 . 2408, 48 . 2422, 49 . 2429,
50 . 2436, 51 . 2443, 52 . 2457, 53 . 2464, 54 . 2471, 55 . 2478, 56 . 2492, 57 . 2499, 58 . 2506, 59 . 2513,
60 . 2527, 61 . 2534, 62 . 2541, 63 . 2548, 64 . 2562, 65 . 2569, 66 . 2576, 67 . 2583, 68 . 2597, 69 . 2604,
70 . 2611, 71 . 2618, 72 . 2632, 73 . 2639, 74 . 2646, 75 . 2653, 76 . 2667, 77 . 2674, 78 . 2681, 79 . 2688,
80 . 2702, 81 . 2709, 82 . 2716, 83 . 2723, 84 . 2737, 85 . 2744, 86 . 2751, 87 . 2758, 88 . 2772, 89 . 2779,
90 . 2786, 91 . 2793, 92 . 2807, 93 . 2814, 94 . 2821, 95 . 2828, 96 . 2842, 97 . 2849, 98 . 2856, 99 . 2863,
100 . 2877, 101 . 2884, 102 . 2891, 103 . 2898, 104 . 2912, 105 . 2919, 106 . 2926, 107 . 2933, 108 . 2947, 109 . 2954,
110 . 2961, 111 . 2968, 112 . 2982, 113 . 2989, 114 . 2996, 115 . 3003, 116 . 3017, 117 . 3024, 118 . 3031, 119 . 3038,
120 . 3052, 121 . 3059, 122 . 3066, 123 . 3073, 124 . 3087, 125 . 3094, 126 . 3101, 127 . 3108, 128 . 3122, 129 . 3129,
130 . 3136, 131 . 3143, 132 . 3157, 133 . 3164, 134 . 3171, 135 . 3178, 136 . 3192, 137 . 3199,
```

2. Python's Tuples are immutable (cannot be changed). Tuples are created using parenthesis (). Create a tuple using the following python code.

```
t = (2,4,6,8,10)
```

Make the following changes to tuple 't'.

- Change the number 4 to 40
- Add another number 12 to this tuple

In the end, the tuple 't' should look as follows.

```
t = (2,40,6,8,10,12)
```

Hint: Since Python tuples are immutable, convert tuple 't' to a list and edit the list. Next convert the new modified list to a tuple.

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3. With a given integral number n , write a program to generate a dictionary that contains $(i, i*i)$ such that i is an integral number between 1 and n (both included). and then the program should print the dictionary.

Suppose the following input is supplied to the program:

8

Then, the output should be:

{1: 1, 2: 4, 3: 9, 4: 16, 5: 25, 6: 36, 7: 49, 8: 64}

4. In HW03 we evaluated the following expressions in Python.

Correct Answer

$$x = 5 + 8i$$

$$y = -6 + 7i$$

$$a) \quad u = x + y$$

$$b) \quad v = xy$$

$$c) \quad w = x / y$$

$$d) \quad z = e^x$$

$$e) \quad r = \sqrt{y}$$

$$f) \quad s = xy^2$$

```
u = (-1+15j)
v = (-86-13j)
w = (0.3058823529411765-0.9764705882352942j)
z = (-21.594119667068036+146.83378286556217j)
r = (1.2687679963832803+2.758581560992251j)
s = (607-524j)
```

The output generated by the problem solved in HW03 was not formatted properly. Repeat the same problem and generate formatted output.

All numbers in the output generated by the problem should have 6 digits after the decimal point. Your output should look as follows.

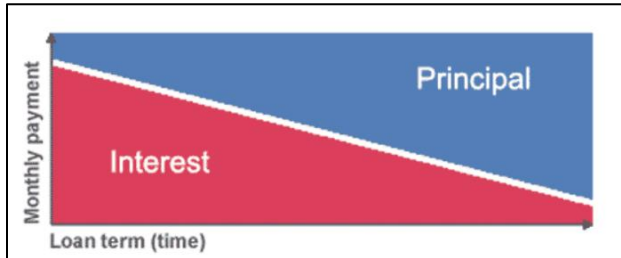
```
u: Real = -1.000000, Imaginary = 15.000000
v: Real = -86.000000, Imaginary = -13.000000
w: Real = 0.305882, Imaginary = -0.976471
z: Real = -21.594120, Imaginary = 146.833783
r: Real = 1.268768, Imaginary = 2.758582
s: Real = 607.000000, Imaginary = -524.000000
```

5. Simulate the mortgage application in Python.

The monthly payment on a loan amount is computed using the following formula.

$$P_{monthly} = \frac{iL}{(1 - (1 + i)^{-m})}$$

Throughout the duration of the loan, the monthly payment remains constant. The principal paid towards the loan increases with time and the monthly interest paid decreases with time.



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Example:

- Loan Amount = $L = 100,000$
- Duration of loan in months = $m = 30 \text{ years} * 12 = 360 \text{ months}$
- Annual interest rate = $I\% = 6\%$
- Monthly interest rate ' i ' = $= \frac{I}{1200} = \frac{6}{1200} = 0.005$

$$t = \left(1 + \frac{6}{1200}\right)^{-360} = 0.166$$

$$P_{monthly} = \frac{\frac{6}{1200} * 100000}{(1 - t)} = \$599.55$$

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Suppose the following information is given:

- Loan amount: \$800,000
- Yearly interest rate = 5%
- Loan duration: 30 years

Write Python code to compute the following:

- Monthly payment
- Total interest amount paid by a customer in 30 years
- Verify that the loan balance is zero after 30 years

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Expected answer:

- Monthly payment = \$4,294.57
- Total interest amount paid by a customer in 30 years = \$746,046.27